MTL 704 April - 10

DFP method - rank 2 operation
quasi newton update method

$B_{k+1} = B_k + \alpha u u^T + \beta v v^T \qquad u \neq 0, v \neq 0$

$B_{k+1} q_k = p_k$ quasi - Newton condition

$$B_{k+1} = B_k + \frac{p_k p_k^T}{p_k^T q_k} - \frac{B_k q_k q_k^T B_k}{q_k^T B_k q_k}$$

B_{k+1} is sym + PD given B_k is sym + PD
(theoritically may not numerically).

$d_k = -B_k \nabla f(x_k) \quad , \quad x_{k+1} = x_k + \alpha_k d_k$

$p_k = x_{k+1} - x_k$

$q_k = \nabla f(x_{k+1}) - \nabla f(x_k)$

look @ mirror reflection / sense of dual

As B_{k+1} is symmetric PD so invertible
Quasi Newton condition

$q_k = B_{k+1}^{-1} p_k$

$\therefore$ we get $q_k = G_{k+1} p_k$

inverse of sym + PD is sym + PD

$\therefore$ $\boxed{G_{k+1} \text{ is sym + PD}}$

rank-2 update will be

$$G_{k+1} = G_k + \frac{q_k q_k^T}{q_k^T p_k} - \frac{G_k p_k p_k^T G_k}{p_k^T G_k p_k}$$

[swaping p_k and q_k in B_{k+1} formula]

$$d_{k+1} = - G_{k+1}^{-1} \nabla f(x_{k+1})$$

Avoid this inverse

- solve system of linear Eq
- cholesky decomposition (sym + PD)
- $d_{k+1} = - B_{k+1} \nabla f(x_{k+1})$

$$B_{k+1} \cdot G_{k+1} = I$$

$$\bar{A} = A + \alpha a a^T + \beta b b^T \quad (a \neq 0, b \neq 0)$$

$$(\bar{A})^{-1} = ?$$

Sherman Morrison Woodbury formula

if $\bar{M} = M + ab^T$ $\quad a \neq 0$, $b \neq 0$

then $$(\bar{M})^{-1} = M^{-1} - \frac{M^{-1} a b^T M^{-1}}{1 + b^T M^{-1} a}$$

$$\therefore \; G_{k+1}^{-1} = B_{k+1} = G_k^{-1} - \frac{G_k^{-1}}{}$$

$$= B_k + \left(1 + \frac{q_k^T B_k q_k}{p}\right)$$

PTO

$$G_{k+1}^{-1} = B_{k+1} = B_k + \left(1 + \frac{q_k^T B_k q_k}{p_k^T q_k}\right) \cdot \frac{p_k p_k^T}{p_k^T q_k}$$

$$- \left(\frac{p_k q_k^T B_k + B_k q_k p_k^T}{p_k^T \; q_k}\right)$$

$\hookrightarrow$ BFGS method of updation

Algo:

$x_0 \in \mathbb{R}^n$, $\varepsilon > 0$

$B_0 : n \times n$
sym pd

Stoping condition: $\|\nabla f(x_k)\| < \varepsilon$

x_k : current iterate

$d_k = -B_k \nabla f(x_k)$

$x_{k+1} = x_k + \alpha_k d_k$

$\uparrow$ usual exact or line search

$p_k = x_{k+1} - x_k$

$q_k = \nabla f(x_{k+1}) - \nabla f(x_k)$

update $B_{k+1} = B_k + \left(1 + \frac{q_k^T B_k q_k}{p_k^T q_k}\right)$ ——

(above formula)

we have preserved $d_k^T \nabla f(x_k) < 0$ theoritically.

→ If this becomes sufficiently large, we can re-start the algo.

Direction
Adaptive ~~Dimention~~ with added momentum

— ADAM

$$\mathcal{L}(x;\theta) = \sum \mathcal{L}_i$$

overall loss function — data — hyper parameters — sum of different loss fun^n

Constrained Optimization

min $f(x)$

subject to: $\left.\begin{array}{l} Ax = b \\ x \geq 0 \end{array}\right] \to$ polyhedron

with $f \in C^1$ and f is convex

$S = \{ x \in \mathbb{R}^n : Ax = b, x \geq 0 \} \to$ non empty closed & bdd. (compact)

is called feasible set of P

$A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ given.

Frank – Wolfe – method

in neighborhood of x_0, estimate f as a linear function (taylor's th) and minimize that. That is linear programming.

First choose initial pt $x_0 \in S$
by solving $\begin{pmatrix} Ax = b \\ x \geq 0 \end{pmatrix}$ and get one x_0

$$\left.\begin{array}{ll} \text{minimize} & 0^T x \quad \text{(zero objective func}^n\text{)} \\ \text{st} & Ax = b \\ & x \geq 0 \end{array}\right\} \text{linear program}$$

this will give 1 solution

use taylor's Approximation

$$f(x) \simeq f(x_0) + (x - x_0)^T Df(x_0)$$
$$= \underbrace{f(x_0) - x_0^T Df(x_0)}_{\substack{\text{known} \\ \therefore \text{ constant}}} + x^T Df(x_0)$$

$\therefore$ don't affect minimization

replace problem (P) by linear approximation P_0 :

$$\left.\begin{array}{ll} \text{minimize} : & Df(x_0)^T x \\ \text{Subject to:} & Ax = b \\ & x \geq 0 \end{array}\right\} \text{Linear program}$$

This is a linear program and can be solved. Let $\bar{x}_0$ be its sol.

set $d_0 = \bar{x}_0 - x_0$

$$\phi(\alpha) = f(x_0 + \alpha d_0) = f(x_0 + \alpha(\bar{x}_0 - x_0))$$
$$= f(\underbrace{(1-\alpha)x_0 + \alpha \bar{x}_0}_{})$$

convex combination of x_0 and $\bar{x}_0$ in S provided $0 \leq \alpha \leq 1$ and we know S is a convex set.

$\therefore$ we solve

$$\min_{0 < \alpha \leq 1} \phi(\alpha) \longrightarrow \alpha_0$$

$$\therefore \quad x_1 = x_0 + \alpha_0 d_0$$

$$= (1-\alpha_0) x_0 + \alpha_0 \bar{x}_0 \in S$$

continue till we get stopping signal

<u>Note:</u> $\quad d_k^T \nabla f(x_k) < 0$

x_k is current pt $\quad \bar{x}_k$ soln of $\min_{(x \in S)} \nabla f(x_k)^T x = w(x)$

and $d_k = \bar{x}_k - x_k$

as $\quad w(\bar{x}_k) \leq w(x) \quad \forall x \in S$

$$\Rightarrow w(\bar{x}_k) \leq w(x_k)$$

$$\Rightarrow \nabla f(x_k)^T \bar{x}_k \leq \nabla f(x_k)^T x_k$$

$$\Rightarrow \nabla f(x_k)^T d_k \leq 0$$

$$\nabla f(x_k)^T d_k < 0 \qquad\qquad \nabla f(x_k)^T d_k = 0$$

$$\Updownarrow \qquad\qquad\qquad\qquad \Updownarrow$$

$$w(\bar{x}_k) < w(x_k) \qquad\qquad w(\bar{x}_k) = w(x_k)$$

current pt.

<u>case</u> $\quad w(x_k) = w(\bar{x}_k)$

i.e. $\quad \nabla f(x_k)^T d_k = 0$

claim: Stop and x_k is the optimal soln of P case $\omega(\bar{x}_k) < \omega(x_k)$

i.e. $\nabla f(x_k)^T d_k < 0$

Do not stop, compute $x_{k+1} = x_k + \alpha_k d_k$

$d_k = \bar{x}_k - x_k$

Justification

Note that (P) is a convex program and in case of $\nabla f(x_k)^T d_k = 0$ then x_k is a KKT point of (P)

By sufficient KKT point x_k is optimal soln of (P) $\therefore$ stop

$$P_k : \min \nabla f(x_k)^T x$$

$$\text{ST} \quad Ax = b \quad \leftarrow \lambda$$

$$x \geq 0 \quad \leftarrow \mu$$

lagrange multiplier

$\bar{x}_k$ is a soln of linear prog. (P_k)

hence convex prog. P_k

$\therefore \bar{x}_k$ is KKT pt. for (P_k)

$\lambda_i \in \mathbb{R}$

$\mu_j \geq 0$

$$L = \nabla f(x_k)^T x + \lambda^T (Ax - b) - \mu^T x$$

lagrange funcn

$\lambda \in \mathbb{R}^m$

$\mu \in \mathbb{R}^n$

$\mu \geq 0$

$$\nabla_x L = 0$$

$$\Rightarrow \nabla f(x_k) + (\lambda^T A)^T - \mu = 0$$

$$\Rightarrow \nabla f(x_k) + A^T \lambda - \mu = 0$$

$$\left.\begin{aligned} \lambda^T (A \bar{x}_k - b) &= 0 \\ \mu^T \bar{x}_k &= 0 \end{aligned}\right] \text{ complementry slackness}$$

$$A \bar{x}_n = b \quad , \quad \bar{x}_k \geq 0 \ , \ \mu \geq 0$$